

```
import streamlit as st
import pandas as pd
import numpy as np
import pickle
import os

st.set_page_config(page_title="Smart Hospital Patient Navigator", page_icon="🏥", layout="wide")
st.markdown(""" """, unsafe_allow_html=True)
@st.cache_resource
def load_model():
    with open('hospital_model.pkl', 'rb') as f:
        return pickle.load(f)
bundle = load_model()
model = bundle['model']
scaler = bundle['scaler']
features = bundle['features']
cols_to_scale = bundle['cols_to_scale']
dept_map_inv = bundle['dept_map_inv']
gender_map = bundle['gender_map']
temp_map = bundle['temp_map']
hr_map = bundle['hr_map']
dur_map = bundle['dur_map']
cc_map = bundle['cc_map']
DEPT_INFO = {
    'Respiratory Medicine': {
        'icon': '🫁', 'color': '#0284c7', 'bg': '#e0f2fe', 'border': '#7dd3fc', 'desc': 'Specialises in conditions affecting the lungs and airways.',
        'next': ['Visit Level 2, Wing B', 'Estimated wait: 15–25 min', 'Please wear a mask']
    },
    'Cardiology': {
        'icon': '❤️', 'color': '#dc2626', 'bg': '#fee2e2', 'border': '#fca5a5', 'desc': 'Specialises in heart and cardiovascular conditions.',
        'next': ['Visit Level 3, Wing A', 'Estimated wait: 20–30 min', 'Bring any previous ECG reports']
    },
    'Gastroenterology': {
        'icon': '🦠', 'color': '#d97706', 'bg': '#fef3c7', 'border': '#fcd34d', 'desc': 'Specialises in digestive system and abdominal conditions.',
        'next': ['Visit Level 1, Wing C', 'Estimated wait: 10–20 min', 'Avoid eating before consultation']
    },
    'Neurology': {
        'icon': '🧠', 'color': '#7c3a21', 'bg': '#fde9e9', 'border': '#f4cccc', 'desc': 'Specialises in brain, spine, and nervous system conditions.',
        'next': ['Visit Level 4, Wing A', 'Estimated wait: 25–35 min', 'Bring list of current medications']
    },
    'General Medicine': {
        'icon': '👨‍⚕️', 'color': '#059669', 'bg': '#d1f2e5', 'border': '#c8e6c9', 'desc': 'Handles general health concerns and non-specialist conditions.',
        'next': ['Visit Level 1, Wing A', 'Estimated wait: 10–15 min', 'Registration desk is open 24/7']
    },
    'Dermatology': {
        'icon': '🧴', 'color': '#b45309', 'bg': '#fff2cc', 'border': '#fde68a', 'desc': 'Specialises in skin, hair, and nail conditions.',
        'next': ['Visit Level 2, Wing D', 'Estimated wait: 15–20 min', 'Bring photos of affected area if possible']
    }
}
# — Hero Header
```

```
st.markdown("""
```

🏥 FUTURE CLASSROOM · MACHINE LEARNING

Smart Hospital Patient Navigator

Find the Right Department for Your Symptoms

```
""", unsafe_allow_html=True) # — Form
```

```
with st.form("triage_form"): # Section 1 — Symptoms
    st.markdown("""
```

1 **What are your main symptoms?** *select all that apply*

```
""", unsafe_allow_html=True)
c1, c2, c3, c4 = st.columns(4)
with c1:
    fever = st.checkbox("🌡️ Fever")
    cough = st.checkbox("🤧 Cough")
with c2:
    headache = st.checkbox("🤕 Headache")
    chest_pain = st.checkbox("❤️ Chest Pain")
with c3:
    stomach_pain = st.checkbox("🤢 Stomach Pain")
    shortness_breath = st.checkbox("😓 Shortness of Breath")
with c4:
    nausea_vomiting = st.checkbox("🤮 Nausea / Vomiting")
    dizziness = st.checkbox("🌀 Dizziness")
c5, _, _, _ = st.columns(4)
with c5:
    skin_rash = st.checkbox("🔴 Skin Rash")
st.markdown("""
```

```
", unsafe_allow_html=True) # Section 2 — Duration & Complaint
st.markdown("""
```

2 **How long have you had these symptoms?**

```
""", unsafe_allow_html=True)
col_cc, col_dur = st.columns(2)
with col_cc:
    chief_complaint = st.selectbox("Chief complaint", options=list(cc_map.keys()))
with col_dur:
    duration = st.selectbox("Duration", options=list(dur_map.keys()), index=1)
st.markdown("""
```

```
", unsafe_allow_html=True) # Section 3 — Severity
st.markdown("""
```

3 **How would you rate the severity?**

```

""" , unsafe_allow_html=True) col_temp, col_hr = st.columns(2) with col_temp: temperature_level =
st.selectbox("Temperature", options=list(temp_map.keys()), index=1) with col_hr: heart_rate_level =
st.selectbox("Heart rate", options=list(hr_map.keys()), index=1) st.markdown("
", unsafe_allow_html=True) # Section 4 — Medical History st.markdown("""

```

4 Do you have any of the following?

```

""" , unsafe_allow_html=True) ch1, ch2, ch3, _ = st.columns(4) with ch1: hypertension = st.checkbox("🩺 High
Blood Pressure") with ch2: heart_disease = st.checkbox("❤️ Heart Disease") with ch3: asthma =
st.checkbox("🤧 Asthma") st.markdown("
", unsafe_allow_html=True) # Section 5 — Patient Info st.markdown("""

```

5 Patient Information

```

""" , unsafe_allow_html=True) col_age, col_gen = st.columns(2) with col_age: age = st.number_input("Age",
min_value=1, max_value=120, value=35) with col_gen: gender = st.selectbox("Gender", options=['Female',
'Male']) submitted = st.form_submit_button("Get AI Recommendation →") # — Result
submitted: patient = pd.DataFrame([{'age': age, 'gender': gender_map.get(gender, 0), 'fever': int(fever),
'cough': int(cough), 'headache': int(headache), 'chest_pain': int(chest_pain), 'stomach_pain':
int(stomach_pain), 'shortness_breath': int(shortness_breath), 'nausea_vomiting': int(nausea_vomiting),
'dizziness': int(dizziness), 'skin_rash': int(skin_rash), 'temperature_level': temp_map.get(temperature_level,
1), 'heart_rate_level': hr_map.get(heart_rate_level, 1), 'duration': dur_map.get(duration, 1), 'asthma':
int(asthma), 'hypertension': int(hypertension), 'heart_disease': int(heart_disease), 'chief_complaint':
cc_map.get(chief_complaint, 9) }) patient_scaled = patient.copy() patient_scaled[cols_to_scale] =
scaler.transform(patient[cols_to_scale]) pred = model.predict(patient_scaled[features])[0] proba =
model.predict_proba(patient_scaled[features])[0] dept_name = dept_map_inv[pred] confidence = proba[pred] *
100 info = DEPT_INFO[dept_name] st.markdown("---") st.markdown("""

```

AI Recommendation

Based on the information you provided

```

""" , unsafe_allow_html=True) res_col, prob_col = st.columns([3, 2]) with res_col: steps_html = ".join( f
' f ' {step}
' for step in info['next'] ) st.markdown(f"""

```

{info['icon']}

{dept_name}

Our AI suggests you visit the {dept_name} Department.

WHY?

{info['desc']} Your reported symptoms and vitals match patients typically directed to this department.

WHAT TO DO NEXT?

{steps_html}

⚠️ This is an AI suggestion, not a medical diagnosis. Please consult a doctor for further evaluation.

```

""" , unsafe_allow_html=True) with prob_col: st.markdown(f"""

```

Confidence by department

```

""" , unsafe_allow_html=True) sorted_depts = sorted(dept_map_inv.items(), key=lambda x: proba[x[0]],
reverse=True) bars_html = "" for idx, dname in sorted_depts: pct = proba[idx] * 100 dinfo =

```

```
DEPT_INFO[dname] is_top = dname == dept_name bars_html += f"""  
{dinfo['icon']} {dname}
```

```
{pct:.1f}%
```

```
""" st.markdown(bars_html + """
```

Model: KNN (k=7) · 102,000 patients · 99.5% accuracy
Powered by: Future Classroom ML

```
""", unsafe_allow_html=True)
```